Machine Protection Door

ASSA ABLOY RP400

ASSA ABLOY
Entrance Systems

Experience a safer
and more open world





HIGH SPEED

Rapid opening and closing ensures short cycle time.

HIGH RELIABILITY

High reliability and low service even after years of high numbers of cycles.

HIGH SAFETY

High level of safety thanks to the safety contact edge with door line photocell; optional a pre-running photocell or integrated light curtain.

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Machine Protection Door ASSA ABLOY RP400

Advantages of the door

- Rapid opening and closing ensures short cycle times up to 5 cycles per minute
- High reliability even after years of high numbers of cycles
- Safety limit switches according to EN ISO 13849-1 and EN 62061
- Doors conform to EN ISO 12100 and EN ISO 14119
- Versions conforming to various factory standards are possible
- Type tested machine protection door with fabric door blade

Design

Side Frames
The side frames are made of aluminum profiles. The front cover side frame can easily be removed and allows easy access for inspection and maintenance.

Windows
For process monitoring when the door is closed, various window variants are available. Available on request is an Anti-UV windows, conform to EN1598 as „welding windows“. Also available is a B1 window „flame resistant“.

Drive Unit
Gear motor with safety gear box including catching device, mounted on the right or left side.

Control System
The door is equipped with a frequency converter control system MCC. The powerful MCC frequency converter control system with Vectoring of the drive unit, connected to the wide range of functions, enables the complex application of the door.

Upon request the door can also be delivered without a control system. Extension box: Space saving extension box for MCC allows mounting of interface card or safety monitoring module for safety limit switches.

Safety Limit Switch integrated in Side Frame
For the indication “door safely closed”, safety limit switches are integrated in the side frame. Optional evaluation by safety systems of the machine (version without safety monitoring module). Due to the use of safety limit switches, machine protection doors cannot be equipped with a knock-out mechanism.

CE Guidelines
This door is designed according to the regulations of the Workplace Directive of the UVV as well as the harmonized CE Guidelines including the EN 13241-1 and 2006/42/EG.

Safety Features
The bottom profile is monitored by an electrical safety contact edge. An optional integrated light curtain covers the full width and height of the doorway, so if anything is detected, the door’s closing movement is stopped immediately and opened to prevent any crashes or injuries. The stationary door line photocell is omitted here. A pre running photo cell is also available as option.

Integrated Cable Chain
Integrated cable chain guides cables for pre-running photocell or electrical safety contact edge inside the side frame.

Indication of Risk
The security of the door is designed for normal use in industrial environments corresponding to the harmonized CE-Guidelines. In addition, special environmental conditions may impact the type of door necessary. For clarification and details, please contact your local sales engineer.

Door Curtain
From partially transparent PVC curtains with colored reinforcement stripes with a high degree of transparency to highly rigid RollTex® and NomaTex®. As well a flame retardant door curtain B1 is available. All door curtains are silicon-free. Side-mounted wind tabs reduce inward bending of the door blade.

RP400 (Color code according to RAL color card)	PVC with colored fabric reinforcement strips	PVC with colored fabric reinforcement strips silicone-free	RollTex® Plus**	RollTex® Original**	NomaTex®	Flame retardant door blade B1
Windows	–	–	•	•	•	•
Vision panels	–	–	–	–	•	–
Blue	• 5002	• 5010	✓5002	• 5002	• A 5005*	
Red		• 3002			• 3000	
Orange		• 2011	• 2009	• 2009	• A 2009*	• 2008
Grey	• 7035	• 7037	• 7001	• 7001	• A 7038	
Yellow		• 1021			• 1023	
Green	• 6018					
Beige	• 1014					
White	• 9016	• 9010				
Black		• 9005				

A = Antistatic properties
* = plain surface on side of top roll
** = DIN 18008-4

Technical Data ASSA ABLOY RP400

TECHNICAL DATA	**ASSA ABLOY RP400**
Application	Interior door / Machine Protection Door
Door dimensions (mm) (DW min./max.) (DH min./max.)	 750 / 4000 mm 750 / 3500 mm
Max. area	14 m²
Opening direction	vertical
Wind resistance (EN 12424)	class 1 (max. 11 m²)
Noise emission	< 70 dB A
Temperature range max. (Extended on request) [2]	+10°C – +45°C (0 up to +60°C)"
Surface material	
Side frame	✓ Anodized aluminum
Top roll cover [1]	• Anodized aluminum
Bottom profile	✓ Anodized aluminum
Motor cover	• Anodized aluminum
Safety features	
Electrical safety contact edge	✓
Door line photocell	✓
Pre-running photocell (Door line photocell as option)	•
Integrated light curtain	•
Flush bottom profile	✓
Drop down protection in drive unit	✓
Emergency egress	Crank handle
Equipment	
LED traffic light	•
Traffic light / Flashing alarm light	•
Safety limit switch in side frame	✓ 1 piece / • 2 piece
Window in sideframe, view on safety limit switch	•
Drive unit	
Motor power	0,75 kW
Chain drive	•
Gear heating	•
Control system	✓ MCC
Control system	
Open/Close max. (m/s) [2]	• 2,4 / 1,3 2,4 / 1,7 (pre-running photocell, light curtain)
Cycles per minute max. * exact number of cycles depending on application	5*
Separate display / control	•
Fuse protection by site	10 – 16 A (B or C characteristic or comparable)
Control voltage	24 V DC
Protection	IP 55
Potential-free contacts	•
UL-Version	•
Network connection	
MCC	3L/(N)/PE/380/400/415/440/460/480V (+/-10%), 50/60 Hz Other voltages on request.
	UL-Version: 3L/(N)PE/208/220/230/460/480V; 60Hz

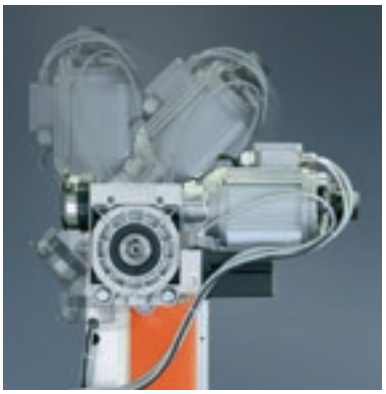
• Option
✓ Standard
*) Specific number of cycles on request



Floor mounting brackets
Option: The door can be mounted freestanding on floor with door support. Levelling screws allow adjustment by uneven floor.

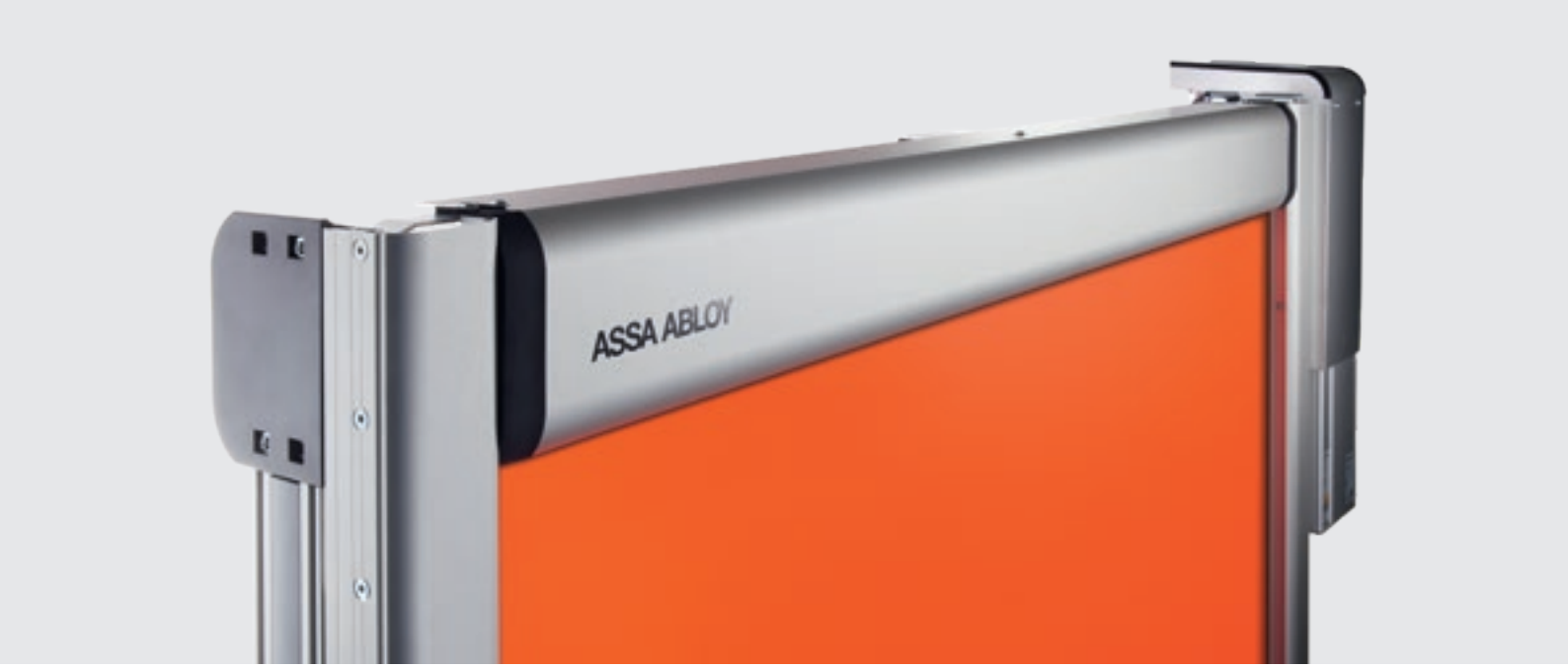


Flush bottom profile
Flush bottom profile without edges to the curtain.



Motor position
Motor can be mounted right- or lefthand side. Motor position can be chosen in step of 45°.
Optional: A chain drive in cramped space requirement for wall mounting only.

1) For doors of height < 2300 mm the use of top roll cover is required according to standard EN 13241-1.
2) Depending on cycles/minute, door size and installation location



Flattened door design. Top roll cover and side frame.

The demands in automated manufacturing processes are high and the smallest interruption or delay can break down the flow of production or endanger employees. Machine protection doors help manufacturers avoid both scenarios. The doors are integrated into the processes and open and close according the production run. They enable the rapid change from complete isolation of the production step to free access to work piece and machine within seconds. It allows short cycle times and can be installed in machines, on production lines and railing systems within close proximity to the machine. Plus, they protect people in highly automated areas, making processes much safer.

Advantages of the ASSA ABLOY RP400

- Service friendly: Two-pieced aluminum side frames offer easy access for inspection and maintenance
- Optional a pre-running photo cell or an integrated light curtain is available for monitoring of the bottom profile. They provide more comfort and higher closing speed
- Integrated cable chain: The cables of the electrical safety contact edge and pre-running photocell are guided in cable chains inside the side frame
- Pluggable cables in the cable chain
- Door support with top roll turned backwards for space saving in front of and behind the door.
- A chain drive for the drive unit is available as option in case of space limitations (only for wall mounting)

Connection with the machine

The in- and outputs of the door control unit MCC can be connected directly with 24 V DC or over potential free contacts to the machine. As standard the following signals are available:

- 2 potential-free safety contacts, when the door is closed (safety limit switch)
- 1 output or potential-free signal, when the door is open
- 1 output or potential-free signal when the door is closed
- 1 output or potential-free signal for faults
- Different fieldbus systems are available as option

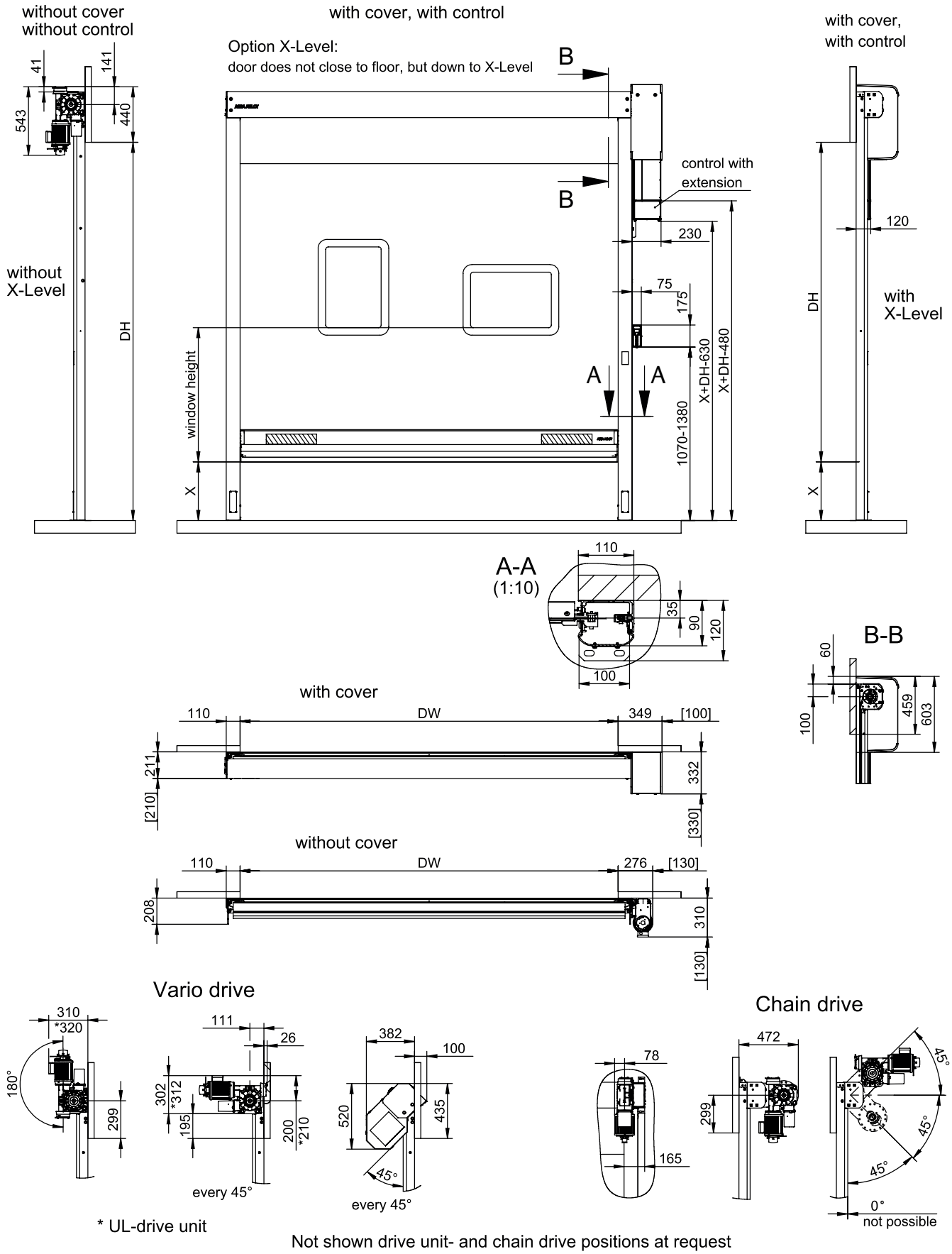


Flat motor cover of the door version free standing.



Completely freestanding installation on the floor – with door support.

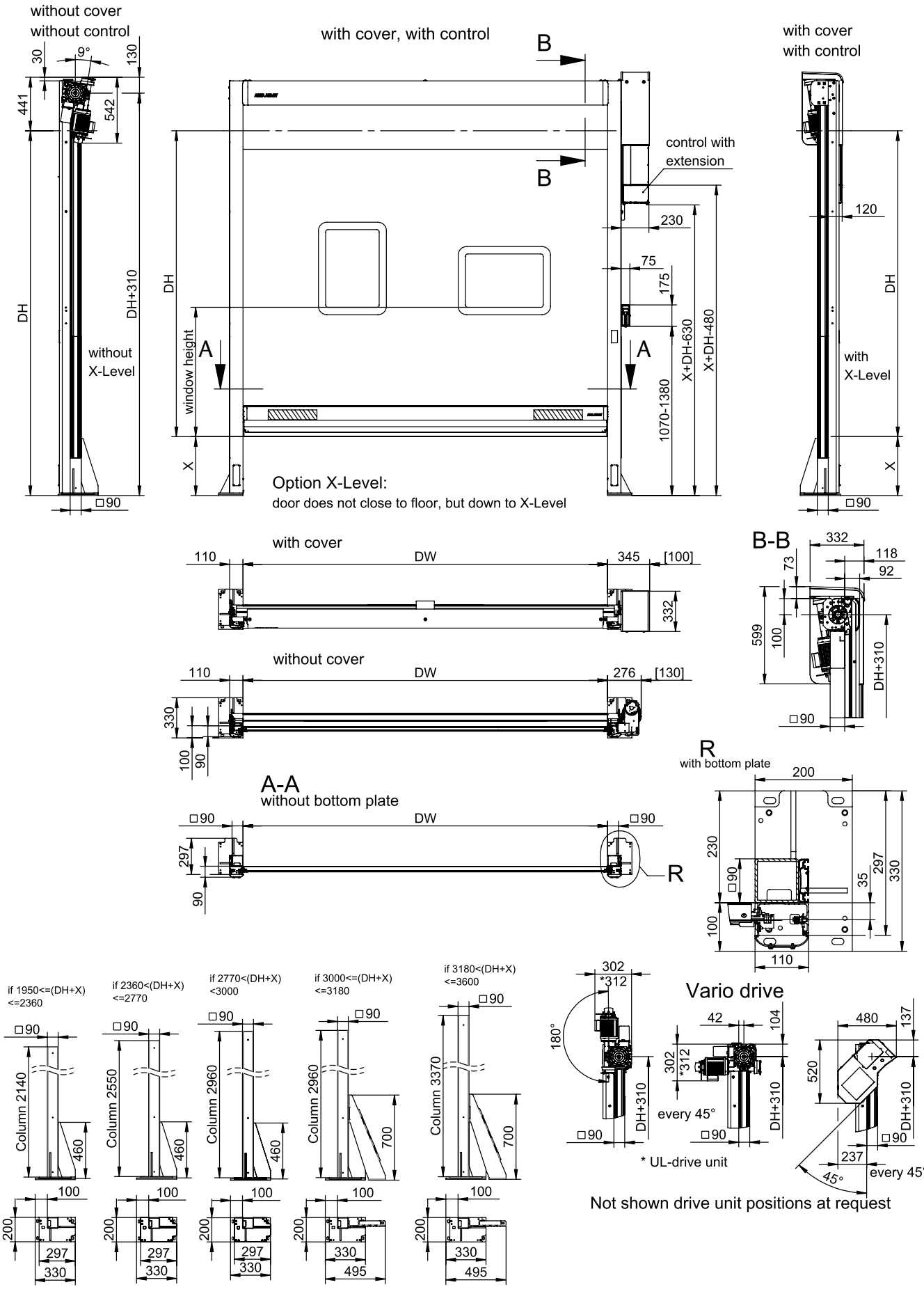
General drawing ASSA ABLOY RP400 no door support



Tolerance ±3 mm (for all dimensions)
[ ] = Space needed for installation

8300R0036

General drawing ASSA ABLOY RP400 with door support



Tolerance ±3 mm (for all dimensions)
[ ] = Space needed for installation

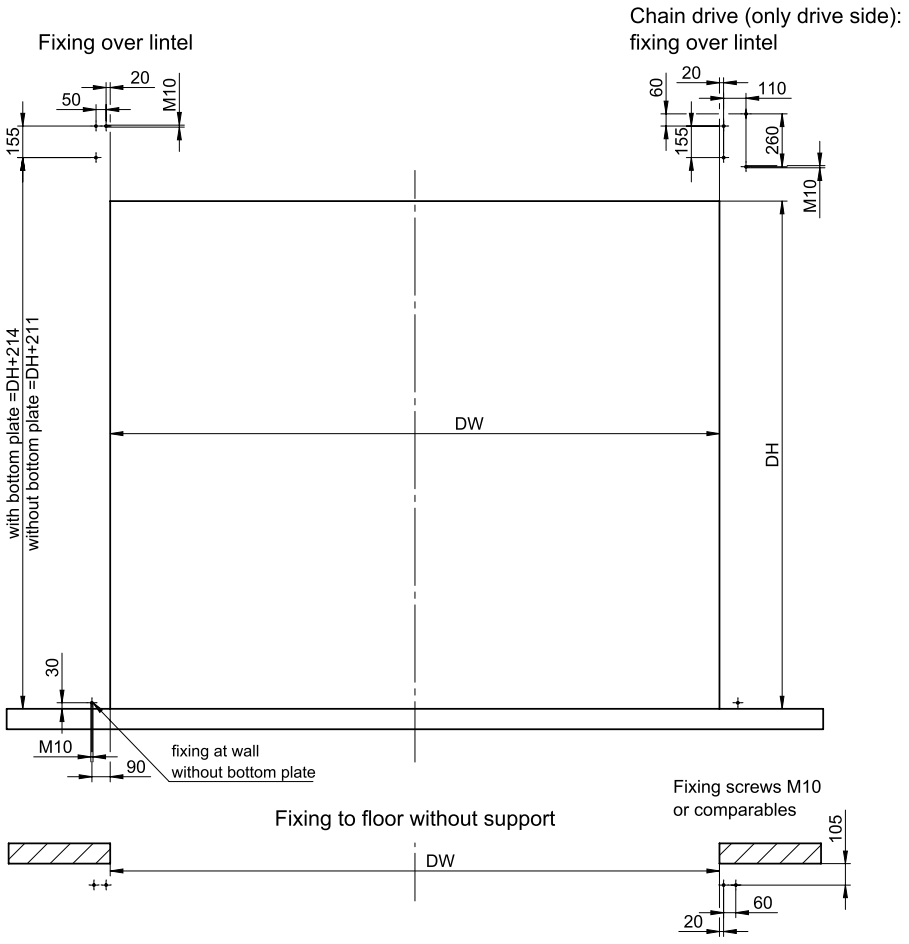
The ASSA ABLOY Group is the global leader in access solutions. Every day, we help billions of people to experience a more open world.

ASSA ABLOY Entrance Systems provides solutions for efficient and safe flow of goods and people. Our offering includes a wide range of automated pedestrian, industrial and residential doors, loading dock equipment, perimeter fencing and service.

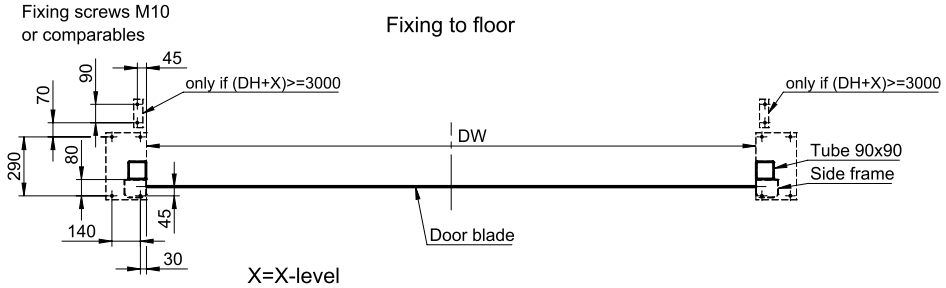
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Fixing plan ASSA ABLOY RP400

Fixing plan RP400 without support



Fixing plan RP400 with support



8300R0037

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